Nifedipine: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Nifedipine: Patient drug information" and "Nifedipine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Procardia XL

Brand Names: Canada

- Adalat XL;
- AG-Nifedipine ER;
- DOM-NIFEdipine [DSC];
- JAMP-Nifedipine ER;
- M-Nifedipine ER;
- MYLAN-NIFEdipine;
- PMS-NIFEdipine ER;
- PMSC-Nifedipine ER;
- PRO-Nifedipine ER

Pharmacologic Category

- Antianginal Agent;
- Antihypertensive;
- Calcium Channel Blocker;
- Calcium Channel Blocker, Dihydropyridine

Dosing: Adult

Dosage guidance:

Safety: The use of IR sublingual nifedipine is **not recommended** for any indication due to safety concerns.

Dosage form information: When switching from IR (oral administration) to ER formulations, use same total daily dose.

Anal fissure

Anal fissure (off-label use):

Note: Administered topically as a local vasodilator in conjunction with supportive measures. Ointment and gel are not commercially available and must be prepared by a licensed compounding facility (Ref).

Peri-anal: 0.2% to 0.3% ointment or gel: After cleansing, apply around fissure(s) 2 to 4 times daily for 4 weeks (Ref).

Angina

Angina:

Chronic stable angina (alternative agent):

Note: A beta-blocker is the preferred initial therapy; if there are ongoing symptoms on beta-blocker therapy, a long-acting dihydropyridine calcium channel blocker (eg, ER nifedipine) may be added; ER nifedipine may also be used as an alternative therapy if there are contraindications or unacceptable adverse effects with beta-blockade (Ref). Use of immediate-release nifedipine (oral or sublingual) is not recommended due to increased adverse effects (hypotension and reflex tachycardia), particularly in absence of a beta-blocker.

Oral: Extended release: Initial: 30 or 60 mg once daily; increase as needed to effective antianginal dose over 1 to 2 weeks. Doses >90 mg/day are rarely needed; maximum: 120 mg/day.

Vasospastic angina:

Note: May be used alone or in combination with nitrates (Ref). Use of immediate-release nifedipine (oral or sublingual) is not recommended due to increased adverse effects (hypotension and reflex tachycardia).

Oral: Extended release: Initial: 30 or 60 mg once daily; increase as needed to effective antianginal dose over 1 to 2 weeks. Doses >90 mg/day are rarely needed; maximum: 120 mg/day.

High-altitude pulmonary edema

High-altitude pulmonary edema (adjunctive therapy) (off-label use):

Prevention: **Note:** May use as an adjunct to gradual ascent in high-risk individuals (eg, history of high-altitude pulmonary edema) (Ref).

Oral: Extended release: 30 mg every 12 hours starting 24 hours prior to ascent; continue for 4 to 5 days after reaching maximal altitude; can extend for up to 7 days in individuals who ascend faster than recommended (Ref).

Treatment: **Note:** Adjunctive to nonpharmacologic measures (eg, oxygen supplementation, portable hyperbaric chamber, gradual descent) or as monotherapy if nonpharmacologic measures are not possible (Ref).

Oral: Extended release: 30 mg every 12 hours; continue until descent is complete, symptoms resolve, and oxygenation is normal for the altitude (Ref).

Hypertension, chronic

Hypertension, chronic:

Note: For patients who warrant combination therapy (eg, BP ≥140/90 mm Hg or suboptimal response to initial monotherapy), may use in combination with another appropriate agent (eg, angiotensin-converting enzyme inhibitor, angiotensin II receptor blocker, thiazide diuretic) (Ref). IR nifedipine (oral or sublingual) should **not** be used for acute BP lowering due to risk of hypotension and ischemic complications. However, may consider extended release for severe asymptomatic hypertension (eg, to lower BP over hours) if there is concern that severe BP elevation will precipitate an acute cardiovascular event, such as in patients with known aortic or intracranial aneurysms (Ref).

Oral: Extended release: Initial: 30 or 60 mg once daily; evaluate response monthly and titrate dose as needed up to 90 mg once daily (Ref); doses up to 120 mg/day are reported (Ref), but if additional BP control is needed in a patient on 90 mg/day, consider combination therapy to reduce risk of adverse effects. Patients with severe asymptomatic hypertension and no signs of acute end organ damage should be evaluated for medication adjustment within 1 week (Ref).

Hypertensive emergency in pregnancy or postpartum

Hypertensive emergency in pregnancy or postpartum (including acute-onset severe hypertension in preeclampsia/eclampsia) (off-label use):

Note: For acute-onset, severe, persistent hypertension (eg, systolic BP ≥160 mm Hg or diastolic BP ≥110 mm Hg) (Ref).

Oral: Extended release: **Note:** Consider for use in less acute settings when slower BP reduction (eg, within 60 minutes versus 5 to 10 minutes) is sufficient (Ref).

Initial: 30 mg; repeat 30 mg after 1 to 2 hours if target BP is not achieved; if systolic or diastolic BP remains above threshold after the second dose, another class of agents should be considered (Ref).

Oral: Immediate release (alternative agent): **Note:** Generally reserved for use when IV access is not available. Some experts avoid use of immediate-release formulations unless immediate BP lowering is warranted (eg, <30 minutes), as it may be associated with precipitous drops in BP (Ref). Do **not** puncture capsule or administer sublingually.

Initial: 10 mg once; if systolic or diastolic BP remains above target at 20 minutes, give a second dose of 10 or 20 mg orally depending on initial response. Reassess BP again 20 minutes after the second dose; if not at target, give a third dose of 10 or 20 mg depending on previous response. If target BP is not achieved 20 minutes after the third dose, another class of agents should be considered; maximum cumulative dose: 50 mg/treatment episode or 180 mg/day (Ref).

Pulmonary arterial hypertension, group 1

Pulmonary arterial hypertension, group 1 (off-label use):

Note: Only used for group 1 pulmonary arterial hypertension patients with a positive vasoreactivity test and under the care of a pulmonary hypertension specialist (Ref).

Oral: Extended release: Initial: 30 mg once daily; titrate gradually, with close hemodynamic monitoring; reported dose range: 120 to 240 mg/day (Ref)

Raynaud phenomenon

Raynaud phenomenon (off-label use):

Oral: Extended release: Initial: 30 mg once daily; if needed, may increase dose gradually, usually once every 4 weeks, but not more frequently than once every 7 to 10 days; monitor blood pressure closely with each dose increase; usual effective dose: 30 to 120 mg/day (Ref).

Tocolysis

Tocolysis (off-label use):

Oral: Immediate release: Initial: 20 to 30 mg as a loading dose, followed by 10 to 20 mg every 3 to 8 hours for up to 48 hours; maximum dose: 180 mg/day (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: Mild to severe impairment: **Oral, Peri-anal (off-label route):** No dosage adjustment necessary (systemic clearance minimally affected by kidney dysfunction); however, use of lower initial doses and more frequent monitoring is recommended in severe impairment (eg, CrCl <30 mL/minute) as these patients are more sensitive to nifedipine's BP lowering effects (Ref).

Hemodialysis, intermittent (thrice weekly): Not significantly dialyzed (Ref): **Oral, Peri-anal (off-label route):** No supplemental dose or dosage adjustment necessary; however, use of lower initial doses and more frequent monitoring is recommended as these patients are more sensitive to nifedipine's BP lowering effects (Ref).

Peritoneal dialysis: Not significantly dialyzed: **Oral, Peri-anal (off-label route):** No dosage adjustment necessary (Ref); however, use of lower initial doses and more frequent monitoring is recommended as these patients are more sensitive to nifedipine's BP lowering effects (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in manufacturer's labeling (has not been studied); use with caution. Clearance of nifedipine is reduced in cirrhotic patients, which may lead to increased systemic exposure; monitor closely for adverse effects/toxicity and consider dose adjustments.

Dosing: Older Adult

Note: Avoid immediate-release formulation (Ref).

Refer to adult dosing. In the management of hypertension, consider lower initial doses and titrate to response (Ref).

Dosing: Pediatric

(For additional information see "Nifedipine: Pediatric drug information")

Hypertension, severe

Hypertension, severe: Limited data available:

Note: Use should be under the supervision of a specialist experienced in the management of pediatric hypertension in the inpatient tertiary setting and used only after other alternatives have been shown to be ineffective. Current pediatric blood pressure guidelines do not recommend nifedipine for the management of acute severe hypertension, as other safe and effective alternatives are available for use in the pediatric population (eg, hydralazine, isradipine) (Ref). If treatment with nifedipine is deemed appropriate and necessary, then administration should occur in the inpatient setting where blood pressure and other hemodynamic parameters can be closely monitored.

Children and Adolescents: Immediate release: Oral: 0.04 to 0.25 mg/kg/dose; maximum single dose: 10 mg/dose; may repeat if needed every 4 to 6 hours; monitor carefully; maximum daily dose: 1 to 2 mg/kg/**day** (Ref).

Hypertension, chronic

Hypertension, chronic: Limited data available: Children and Adolescents (able to swallow whole tablet): Extended release: Oral: Initial: 0.2 to 0.5 mg/kg/**day** once daily or divided in 2 doses every 12 hours; do not exceed initial adult daily dose of 30 to 60 mg/**day**; titrate dose to effect; maximum daily dose: 3 mg/kg/**day** up to 120 mg/**day** (Ref); some centers use a higher maximum dose: 3 mg/kg/**day** up to 180 mg/**day** (Ref). **Note:** In adults, doses are usually titrated upward over 7 to 14 days; may increase over 3 days if clinically necessary (Ref).

High-altitude pulmonary edema, treatment

High-altitude pulmonary edema, treatment: Limited data available: **Note:** Reserve treatment with NIFEdipine for unsatisfactory response to oxygen and/or altitude descent:

Children and Adolescents:

Immediate release: Oral: 0.5 mg/kg/dose every 8 hours; maximum dose: 20 mg/dose (Ref).

Extended release (preferred if weight allows): Oral: 1.5 mg/kg/day once daily or divided in 2 doses per **day**; do not exceed adult dose of 30 mg every 12 hours. **Note:** Smallest extended-release dosage form available is 30 mg (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling (has not been studied); the pharmacokinetics of nifedipine are not significantly influenced by the degree of renal impairment (only trace amounts of unchanged drug are found in urine). Based on experience in adult dialysis patients, supplemental doses are not necessary with hemodialysis or peritoneal dialysis; however, more frequent monitoring may be necessary.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling (has not been studied); use with caution. Clearance of nifedipine is reduced in cirrhotic adult patients, which may lead to increased systemic exposure; monitor closely for adverse effects/toxicity and consider dose adjustments.

Adverse Reactions (Significant): Considerations

Angina/Myocardial infarction

Exacerbation of angina pectoris and/or **acute myocardial infarction** (MI) have occurred with initiation or dosage titration of immediate release (IR) (orally or sublingually) nifedipine. **Hypotension** and reflex **tachycardia** may occur resulting in angina and/or MI in patients with coronary heart disease, especially in the absence of concurrent beta-blockade. Use of IR nifedipine has been associated with increased mortality in patients with coronary heart disease. In patients with unstable angina/non–ST-elevation myocardial infarction (STEMI), the use of IR nifedipine is not recommended in the absence of a beta-blocker and is contraindicated in patients with STEMI. Effects may be reversible upon discontinuation (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Hypotension due to calcium channel blocking properties may cause a reflex increase in sympathetic activity, leading to ischemic effects. In addition, the IR nature of nifedipine may cause more marked fluctuations in hemodynamics compared to extended release (ER) formulations (Ref).

Onset: Varied; increase in mortality observed after 2 weeks (Muller 1984). However, hemodynamic effects (ie, decrease in blood pressure, increase in heart rate, ECG changes) observed as quickly as after 1 dose (Ref).

Risk factors:

- Acute MI (Ref)
- Coronary heart disease (Ref)
- Refractory angina (Ref)

Hypotension/syncope

Symptomatic **hypotension** with or without **syncope** may occur when using the immediate release (IR) formulation. The use of IR nifedipine (sublingually or orally) in hypertensive emergencies and urgencies is neither safe or effective and is associated with increased risk of adverse events such as acute myocardial infarction, cerebrovascular ischemia, fetal distress, and death when used to treat hypertension (Ref). IR nifedipine may be warranted for acute treatment in pregnant or postpartum women based on guidelines (Ref). Although hypotension should be reversible upon discontinuation, effects may be initially severe and resistant to treatment (Ref).

Mechanism: Dose-related; related to the pharmacologic action. Inhibits calcium channels in peripheral smooth muscle leading to vasodilation. IR nifedipine, compared to extended release (ER) nifedipine has a much faster rate of increase in plasma levels, leading to more adverse reactions (Ref).

Onset: Rapid; hemodynamic effects (ie, decrease in blood pressure, increase in heart rate) observed as quickly as after one dose (Ref).

Risk factors:

• Older patients (age 60 to 83 years) (Ref)

Peripheral edema

Peripheral edema is a common adverse reaction of calcium channel blockers, characterized by ankle and leg swelling independent of fluid retention (Ref). It is a bothersome adverse reaction for patients and may lead to discontinuation (Ref). Peripheral edema can be expected to subside within several days following intervention.

Mechanism: Dose- and time-related; related to the pharmacologic action. Calcium channel blocker-mediated peripheral edema is caused by arteriolar vasodilation that subsequently leads to increased hydrostatic pressure in the precapillary circulation and fluid movement from the capillary vasculature to the interstitial space (Ref). In addition, impaired postural vasoconstriction may contribute (Ref).

Onset: Varied; has been reported between 4 weeks to >6 months after initiation (Ref).

Risk factors:

• Dose-related (ie, higher doses) (Ref); however, may develop more frequently and at lower doses in patients with impaired postural autoregulation (eg, diabetes, arterial disease) (Ref).

• Duration-related (>6 months) (Ref)

• Dihydropyridine calcium channel blockers (DHPs) versus non-DHPs (Ref)

• Lipophilic DHPs (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Incidence may include data with concomitant beta-blocker therapy. ER = extended release; IR = immediate release.

>10%:

Cardiovascular: Flushing (IR: 25%; ER: <3%), peripheral edema (4% to 30%) (table 1)Peripheral Edema

Table 1: **Nifedipine: Adverse Reaction: Peripheral Edema**

Drug (Nifedipine)	Placebo	Dose	Dosage Form	Number of Patients (Nifedipine
30%	N/A	180 mg per day	ER tablets	N/A
10%	N/A	N/A	ER tablets	N/A
13%	N/A	120 mg per day or more	IR capsules	N/A
7%	1%	N/A	IR capsules	226
4%	N/A	less than 60 mg per day	IR capsules	N/A

Gastrointestinal: Heartburn (IR: ≤11%), nausea (IR: ≤11%; ER: 3%)

Nervous system: Dizziness (IR: 27%), headache (16% to 23%)

Neuromuscular & skeletal: Asthenia (IR: 12%; ER: <3%)

1% to 10%:

Cardiovascular: Acute myocardial infarction (4%), cardiac failure (2%), palpitations (≤7%), transient hypotension (5%)

Dermatologic: Dermatitis (IR), diaphoresis, pruritus, skin rash (ER), urticaria

Gastrointestinal: Abdominal cramps (IR), abdominal pain (ER), constipation (≤3%), diarrhea, dyspepsia (ER), flatulence, xerostomia (ER)

Genitourinary: Impotence (ER), sexual difficulty (IR)

Nervous system: Balance impairment (IR), chills (IR), drowsiness (ER), fatigue (ER: 6%), insomnia (ER), jitteriness (IR), mood changes (IR: ≤7%), nervousness (≤7%), pain (ER), paresthesia (ER), shakiness (IR), sleep disturbance (IR)

Neuromuscular & skeletal: Arthralgia (ER), joint stiffness (IR), lower limb cramp (ER), muscle cramps (IR: ≤8%), tremor (IR: ≤8%; ER: <1%)

Ophthalmic: Blurred vision (IR)

Renal: Polyuria (ER: <3%; IR: <1%)

Respiratory: Chest congestion (IR), cough (IR: ≤6%; ER: <1%), dyspnea, nasal congestion (IR), pleuritic chest pain (ER), pulmonary edema (2%), wheezing (IR: ≤6%)

Miscellaneous: Fever, inflammation (IR)

<1%:

Cardiovascular: Cardiac arrhythmia, cardiac conduction disorder, erythromelalgia, exacerbation of angina pectoris, facial edema, hypotension (may be severe hypotension), syncope, tachycardia, ventricular arrhythmia

Dermatologic: Alopecia, exfoliative dermatitis

Endocrine & metabolic: Decreased libido, gout, gynecomastia, hot flash, weight gain

Gastrointestinal: Dysgeusia, eructation, gastroesophageal reflux disease, gastrointestinal hemorrhage, gastrointestinal irritation, gingival hyperplasia, melena, vomiting

Genitourinary: Dysuria, hematuria, mastalgia, nocturia

Hematologic & oncologic: Anemia, leukopenia, purpuric disease, thrombocytopenia

Hepatic: Hepatitis (allergic)

Hypersensitivity: Angioedema

Nervous system: Anxiety, ataxia, depression, hypertonia, hypoesthesia, malaise, migraine, nightmares, paranoid ideation, rigors, vertigo

Neuromuscular & skeletal: Arthritis (with positive ANA), back pain, myalgia

Ophthalmic: Abnormal lacrimation, periorbital edema, transient blindness (at peak of plasma level; can be unilateral), visual disturbance

Otic: Tinnitus

Respiratory: Epistaxis, sinusitis, upper respiratory tract infection

Frequency not defined:

Endocrine & metabolic: Increased lactate dehydrogenase

Gastrointestinal: Cholecystitis

Hepatic: Increased serum alanine aminotransferase, increased serum alkaline phosphatase, increased serum aspartate aminotransferase

Neuromuscular & skeletal: Increased creatine phosphokinase in blood specimen

Postmarketing:

Dermatologic: Acute generalized exanthematous pustulosis (Ref), bullous skin disease (Ref), erythema multiforme (Ref), psoriasis (Ref), skin photosensitivity (Ref), Stevens-Johnson syndrome (Ref), toxic epidermal necrolysis

Gastrointestinal: Bezoar formation, gastrointestinal obstruction (Ref), gastrointestinal ulcer (Ref)

Hematologic & oncologic: Malignant neoplasm of lip (Ref), positive direct Coombs test

Contraindications

Hypersensitivity to nifedipine or any component of the formulation.

Note: Considered contraindicated in patients with ST-elevation myocardial infarction (STEMI) (ACCF/AHA [O'Gara 2013]); avoid use (Elkayam 1990; FDA 2015).

Canadian labeling: Additional contraindications (not in US labeling): Hypersensitivity to other dihydropyridine calcium antagonists; severe hypotension; cardiovascular shock; concomitant use with rifampicin; breast-feeding; pregnancy or women of childbearing potential. Note: SOGC guidelines recommend nifedipine as a preferred agent for maternal hypertension (SOGC [Magee 2022]). Extended release only: Kock pouch (ileostomy after proctocolectomy); moderate or severe hepatic impairment; severe gastrointestinal obstructive disorders. Immediate release only: Acute myocardial infarction.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• Hypotension/syncope: **IR nifedipine should not be used for acute BP reduction.**

Disease-related concerns:

• Aortic stenosis: Use with extreme caution in patients with severe aortic stenosis; may reduce coronary perfusion resulting in myocardial ischemia.

• GI strictures: Alterations in GI anatomy (eg, severe GI narrowing, history of GI cancer, obstruction, bowel resection, gastric bypass, vertical banded gastroplasty) and underlying hypomotility disorders have led to bezoar formation with extended release forms.

• Heart failure: The American College of Cardiology/American Heart Association heart failure guidelines recommend to avoid use in patients with heart failure due to lack of benefit and/or worse outcomes with calcium channel blockers in general (FDA 2015).

• Hepatic impairment: Use with caution in patients with hepatic impairment. Clearance of nifedipine is reduced in cirrhotic patients leading to increased systemic exposure; monitor closely for adverse effects/toxicity and consider dose adjustments.

• Hypertrophic cardiomyopathy (HCM) with outflow tract obstruction: Use with caution in patients with HCM and outflow tract obstruction since reduction in afterload may worsen symptoms associated with this condition.

Dosage form specific issues:

• Extended release formulation: Consists of drug within a nondeformable matrix; following drug release/absorption, the matrix/shell is expelled in the stool. The use of nondeformable products in patients with known stricture/narrowing of the GI tract (eg, severe gastrointestinal narrowing, colon cancer, obstruction, bowel resection, gastric bypass, vertical banded gastroplasty) has been associated with symptoms of obstruction (pharmacobezoar).

• Immediate release formulation: Immediate release formulations should not be used to manage primary hypertension, adequate studies to evaluate outcomes have not been conducted.

• Lactose: Adalat CC tablets contain lactose; do not use with galactose intolerance, congenital lactase deficiency, or glucose-galactose malabsorption syndromes.

Other warnings/precautions:

• Surgery: Use with caution before major surgery. Cardiopulmonary bypass, intraoperative blood loss or vasodilating anesthesia may result in severe hypotension and/or increased fluid requirements. Consider withdrawing nifedipine (>36 hours) before surgery if possible.

• Withdrawal: Abrupt withdrawal may cause rebound angina in patients with coronary artery disease.

Warnings: Additional Pediatric Considerations

The ”bite and swallow” method of administering immediate-release nifedipine is not recommended; variability of dose supplied via this method may lead to an overdose and result in significant side effects (eg, hypotension) in pediatric patients (Flynn 2003). An extemporaneous solution should be compounded to provide the most accurate dosing (see ”Extemporaneous Preparations”).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

Generic: 10 mg, 20 mg

Tablet Extended Release 24 Hour, Oral:

Procardia XL: 30 mg, 60 mg, 90 mg [DSC]

Generic: 30 mg, 60 mg, 90 mg

Generic Equivalent Available: US

Yes

Pricing: US

Capsules (NIFEdipine Oral)

10 mg (per each): $0.46 - $1.29

20 mg (per each): $0.99 - $2.30

Tablet, 24-hour (NIFEdipine ER Oral)

30 mg (per each): $0.41 - $1.39

60 mg (per each): $0.54 - $2.48

90 mg (per each): $0.67 - $3.03

Tablet, 24-hour (NIFEdipine ER Osmotic Release Oral)

30 mg (per each): $1.10 - $1.39

60 mg (per each): $1.83 - $3.50

90 mg (per each): $2.31 - $2.91

Tablet, 24-hour (Procardia XL Oral)

30 mg (per each): $7.93

60 mg (per each): $13.72

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer’s AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule, Oral:

Generic: 5 mg, 10 mg

Tablet Extended Release 24 Hour, Oral:

Adalat XL: 30 mg

Generic: 30 mg, 60 mg

Extemporaneous Preparations

4 mg/mL Oral Suspension (ASHP Standard Concentration)

A 4 mg/mL oral suspension may be made with liquid capsules (**Note:** Concentration inside capsule may vary depending on manufacturer. Procardia: 10 mg capsule contains a concentration of 10 mg/0.34 mL [29.4 mg/mL]). Puncture the top of twelve 10 mg liquid capsules with one needle to create a vent. Insert a second needle attached to a syringe and extract the liquid; transfer to a calibrated bottle and add sufficient quantity of a 1:1 mixture of Ora-Sweet and Ora-Plus to make 30 mL. Label ”shake well”. Stable 90 days under refrigeration or at room temperature (Ref).

Administration: Adult

Oral:

Immediate release: In general, may be administered with or without food.

Extended release: Tablets should be swallowed whole; do not crush, split, or chew.

Adalat CC, Afeditab CR: Administer on an empty stomach (per manufacturer). Other extended release products may not have this recommendation; consult product labeling.

Bariatric surgery: Some institutions may have specific protocols that conflict with these recommendations; refer to institutional protocols as appropriate. IR capsule formulation is available. If safety and efficacy of nifedipine can be effectively monitored, no change in formulation or administration is required after bariatric surgery; however, selection of IR formulation or alternative therapy is advised for cardiovascular and other high-risk labeled and off-label indications.

Administration: Pediatric

Oral:

Immediate release: In general, administer with or without food. If using extemporaneously compounded oral suspension, use calibrated device to accurately measure dose.

Extended release: Tablets should be swallowed whole; do not crush, break, chew, or divide.

Storage/Stability

Adalat CC, Afeditab CR, Procardia XL: Store below 30°C (86°F); protect from light and moisture.

Nifedical XL: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F); protect from light and moisture.

Immediate release capsules (Procardia): Store at 15°C to 25°C (59°F to 77°F); prevent capsules from freezing; protect from light and moisture.

Use: Labeled Indications

Angina: Management of chronic stable or vasospastic angina.

Hypertension, chronic: Management of hypertension (ER products only).

Use: Off-Label: Adult

Anal fissure; High-altitude pulmonary edema; Hypertensive emergency in pregnancy or postpartum (including acute-onset severe hypertension in preeclampsia/eclampsia); Pulmonary arterial hypertension, group 1; Raynaud phenomenon; Tocolysis

Medication Safety Issues

Sound-alike/look-alike issues:

NIFEdipine may be confused with niCARdipine, niMODipine, nisoldipine

Procardia XL may be confused with Cartia XT

Older Adult: High-Risk Medication:

Beers Criteria: Nifedipine (immediate release) is identified as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to its potential to cause hypotension and risk for precipitating myocardial ischemia (Beers Criteria [AGS 2023]).

Nifedipine is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) with severe symptomatic aortic stenosis (O'Mahony 2023).

International issues:

Depin [India] may be confused with Depen brand name for penicillamine [US]; Depon brand name for acetaminophen [Greece]; Dipen brand name for diltiazem [Greece]

Nipin [Italy and Singapore] may be confused with Nipent brand name for pentostatin [US, Canada, and multiple international markets]

Metabolism/Transport Effects

Substrate of CYP2D6 (Minor), CYP3A4 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential; **Inhibits** ENT1 and CNT3

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Alcohol (Ethyl): May increase serum concentrations of NIFEdipine (Systemic). *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Atosiban: Calcium Channel Blockers may increase adverse/toxic effects of Atosiban. Specifically, there may be an increased risk for pulmonary edema and/or dyspnea. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta-Blockers: NIFEdipine (Systemic) may increase hypotensive effects of Beta-Blockers. NIFEdipine (Systemic) may increase negative inotropic effects of Beta-Blockers. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Calcium Salts: May decrease therapeutic effects of Calcium Channel Blockers. *Risk C: Monitor*

Cardiac Glycosides: NIFEdipine (Systemic) may increase serum concentrations of Cardiac Glycosides. *Risk C: Monitor*

Cimetidine: May increase serum concentrations of Calcium Channel Blockers. *Risk C: Monitor*

Cladribine: Inhibitors of Equilibrative Nucleoside (ENT1) and Concentrative Nucleoside (CNT3) Transporters may increase serum concentrations of Cladribine. Management: Avoid concomitant use of ENT1 or CNT3 inhibitors during the 4 to 5 day oral cladribine treatment cycles whenever possible. If combined, consider an ENT1 or CNT3 inhibitor dose reduction and separation in the timing of administration. *Risk D: Consider Therapy Modification*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

CloZAPine: May increase hypotensive effects of NIFEdipine (Systemic). NIFEdipine (Systemic) may increase serum concentrations of CloZAPine. *Risk C: Monitor*

CycloSPORINE (Systemic): Calcium Channel Blockers (Dihydropyridine) may increase serum concentrations of CycloSPORINE (Systemic). CycloSPORINE (Systemic) may increase serum concentrations of Calcium Channel Blockers (Dihydropyridine). *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of NIFEdipine (Systemic). *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of NIFEdipine (Systemic). Management: Avoid coadministration of nifedipine with strong CYP3A4 inducers when possible and if combined, monitor patients closely for clinical signs of diminished nifedipine response. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of NIFEdipine (Systemic). *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of NIFEdipine (Systemic). Management: Consider alternatives to this combination when possible. If combined, initiate nifedipine at the lowest dose available and monitor patients closely for increased nifedipine effects and toxicities (eg, hypotension, edema). *Risk D: Consider Therapy Modification*

Dantrolene: May increase hyperkalemic effects of Calcium Channel Blockers. Dantrolene may increase negative inotropic effects of Calcium Channel Blockers. Management: Consider alternatives to the use of dantrolene with calcium channel blockers during the management of malignant hyperthermia when possible, as product labeling states that the concomitant use of these agents is not recommended. *Risk D: Consider Therapy Modification*

Dapoxetine: May increase orthostatic hypotensive effects of Calcium Channel Blockers. *Risk C: Monitor*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

FLUoxetine: May increase serum concentrations of NIFEdipine (Systemic). *Risk C: Monitor*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Grapefruit Juice: May increase serum concentrations of NIFEdipine (Systemic). *Risk X: Avoid*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Inhalational Anesthetics: May increase hypotensive effects of Calcium Channel Blockers. *Risk C: Monitor*

Isocarboxazid: May increase antihypertensive effects of Antihypertensive Agents. *Risk X: Avoid*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Magnesium Sulfate: May increase adverse/toxic effects of Calcium Channel Blockers (Dihydropyridine). Specifically, the risk of hypotension or muscle weakness may be increased. *Risk C: Monitor*

Melatonin: May decrease antihypertensive effects of Calcium Channel Blockers (Dihydropyridine). *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Micafungin: May increase serum concentrations of NIFEdipine (Systemic). *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Neuromuscular-Blocking Agents (Nondepolarizing): Calcium Channel Blockers may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

QuiNIDine: NIFEdipine (Systemic) may decrease serum concentrations of QuiNIDine. NIFEdipine (Systemic) may increase serum concentrations of QuiNIDine. QuiNIDine may increase serum concentrations of NIFEdipine (Systemic). *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Tacrolimus (Systemic): Calcium Channel Blockers (Dihydropyridine) may increase serum concentrations of Tacrolimus (Systemic). *Risk C: Monitor*

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

VinCRIStine: NIFEdipine (Systemic) may increase serum concentrations of VinCRIStine. *Risk C: Monitor*

Food Interactions

Nifedipine serum levels may be decreased if taken with food. Food may decrease the rate but not the extent of absorption of Procardia XL. Increased nifedipine concentrations resulting in therapeutic and vasodilator side effects, including severe hypotension and myocardial ischemia, may occur if nifedipine is taken by patients ingesting grapefruit. Management: Avoid grapefruit/grapefruit juice.

Reproductive Considerations

Medications considered acceptable for the treatment of chronic hypertension during pregnancy may generally be continued in patients trying to conceive. Nifedipine is one of the medications patients can transition to if currently taking an agent that is ineffective or should be avoided during pregnancy (ACC/AHA [Whelton 2018]. ACOG 2019, NICE 2019).

Pregnancy Considerations

Nifedipine crosses the placenta (Mulrenin 2021; van de Vusse 2022).

An increase in perinatal asphyxia, cesarean delivery, prematurity, and intrauterine growth restriction have been reported following maternal use (ESC [Regitz-Zagrosek 2018]).

Chronic maternal hypertension is also associated with adverse events in the fetus/infant. Chronic maternal hypertension may increase the risk of birth defects, low birth weight, premature delivery, stillbirth, and neonatal death. Actual fetal/neonatal risks may be related to the duration and severity of maternal hypertension. Untreated chronic hypertension may also increase the risks of adverse maternal outcomes, including gestational diabetes, preeclampsia, delivery complications, stroke, and myocardial infarction (ACOG 2019a).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of nifedipine may be altered. Close monitoring is recommended (Mulrenin 2021; van de Vusse 2022).

Patients with preexisting hypertension may continue their medication during pregnancy unless contraindications exist (ESC [Regitz-Zagrosek 2018]). When treatment for chronic hypertension is initiated during pregnancy, long-acting oral nifedipine is one of the preferred agents; avoid in patients with tachycardia (ACOG 2019a; ESC [Regitz-Zagrosek 2018]; SOGC [Magee 2022]). IR oral nifedipine is recommended for the management of acute-onset, severe hypertension in pregnant and postpartum patients, including those with preeclampsia or eclampsia (ACOG 2020; ESC [Cífková 2020]; ESC [Regitz-Zagrosek 2018]; SOGC [Magee 2022]).

IR oral nifedipine has also been evaluated for the treatment of preterm labor. Tocolytics may be used for the short-term (48-hour) prolongation of pregnancy to allow for the administration of antenatal steroids and should not be used prior to fetal viability or when the risks of use to the fetus or mother are greater than the risk of preterm birth (ACOG 2016). Nifedipine is ineffective for maintenance tocolytic therapy (ACOG 2016; Aggarwal 2018; Roos 2013; Verspyck 2017).

Breastfeeding Considerations

Nifedipine is present in breast milk.

Multiple reports summarize data related to the presence of nifedipine in breast milk:

• Breast milk was sampled in a patient following a full-term delivery who received nifedipine 10 mg/day prior to the study. Nifedipine 20 mg was administered on the day of the study at 10 days postpartum. Nifedipine and the pyridine metabolite were present in breast milk. Peak concentrations occurred ~1 hour following the maternal dose (Penny 1989).

• Following delivery at 26 weeks gestation, breast milk was sampled in a patient who received nifedipine 30 mg every 8 hours for 48 hours, followed by nifedipine 20 mg every 8 hours for 48 hours, then 10 mg every 8 hours for 36 hours. The highest breast milk concentration (53.35 ng/mL) occurred within 1 hour of the 30 mg dose. The highest breast milk concentrations following the 20 mg and 10 mg doses were 16.35 ng/mL and 12.89 ng/mL, respectively. Nifedipine was measurable in breast milk 8 hours following the 20 mg and 30 mg doses, but not the 10 mg dose. The reported half-life of nifedipine in breast milk was ~3 hours (Ehrenkranz 1989).

• Using a milk concentration of 53.35 ng/mL, the estimated daily infant dose of nifedipine via breast milk is 8 mcg/kg/day providing a relative infant dose (RID) of 0.27% to 3.2% compared to an infant therapeutic dose of 0.25 to 3 mg/kg/day.

• In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000).

• In addition to maternal dose, breast milk concentrations may be influenced by maternal ABCG2 genotype. Nifedipine is a substrate to ABCG2 transporter which is expressed on lactating breasts. Patients genotyped as wild-type homozygous 421CC (n=13) had lower breast milk concentrations of nifedipine than patients genotyped as 421CA heterozygous (n=6) following a maternal dose of slow-release nifedipine 20 mg administered 15 to 30 days postpartum (Malfará 2019).

The use of nifedipine in breastfeeding patients for the treatment of Raynaud phenomenon has been described in case reports and case series; no adverse events were noted in the infants exposed to nifedipine via breast milk (Anderson 2004; Barrett 2013; Garrison 2002; Jansen 2019; Page 2006; Wu 2012).

Nifedipine is considered compatible with breastfeeding (ESC [Cífková 2020]; SOGC [Magee 2022]; WHO 2002), although data following long-term use is limited (WHO 2002). According to the manufacturer, the decision to continue or discontinue breastfeeding during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother.

The Academy of Breastfeeding Medicine recommends the use of nifedipine for the treatment of Raynaud phenomenon of the nipple in breastfeeding mothers (ABM [Berens 2016]).

Dietary Considerations

Avoid grapefruit juice with all products.

Immediate release: Capsule is rapidly absorbed orally if it is administered without food, but may result in vasodilator side effects; if flushing is problematic, administration with low-fat meals may decrease. In general, can take with or without food.

Monitoring Parameters

Heart rate; blood pressure; peripheral edema.

Mechanism of Action

Inhibits calcium ion from entering the “slow channels” or select voltage-sensitive areas of vascular smooth muscle and myocardium during depolarization, producing a relaxation of coronary vascular smooth muscle and coronary vasodilation; increases myocardial oxygen delivery in patients with vasospastic angina; also reduces peripheral vascular resistance, producing a reduction in arterial blood pressure.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Immediate release: ~20 minutes

Protein binding (concentration dependent): 92% to 98%; **Note:** Protein-binding may be significantly decreased in patients with renal or hepatic impairment

Metabolism: Hepatic via CYP3A4 to inactive metabolites

Bioavailability: Capsule: 40% to 77%; ER: 65% to 89% relative to immediate release capsules; bioavailability increased with significant hepatic disease

Half-life elimination: Adults: Healthy: 2 to 5 hours; Cirrhosis: 7 hours; Elderly: 7 hours (extended release tablet)

Excretion: Urine (60% to 80% as inactive metabolites); feces

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: Mean C_{max} is 36% higher and plasma concentration is 70% greater in elderly patients.

Patient Counseling Points

(For additional information see ”Nifedipine: Patient drug information”)

What is this drug used for?

• It is used to treat chest pain or pressure.

• Some forms of this drug are approved to treat high blood pressure.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Fatigue

• Headache

• Flushing

• Heartburn

• Nausea

• Loss of strength and energy

• Anxiety

• Tablet shell in stool

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Severe dizziness

• Passing out

• Chest pain

• Abnormal heartbeat

• Mood changes

• Shortness of breath

• Excessive weight gain

• Swelling of arms or legs

• Muscle pain

• Muscle cramps

• Tremors

• Severe abdominal pain

• Severe constipation

• Black, tarry, or bloody stools

• Vomiting blood

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Adalat | Coracten | Epilat | Fenamon | Nifecard | Zenusin;
- (AR) Argentina: Adalat | Nifecor | Nifed sol | Nifedel | Nifedipina | Nifedipina r 20 | Nifelat | Nifenitron | Nitenk;
- (AT) Austria: Adalat | Fedip | Majolat | Nifebene | Nifedipin stada artzneimittel | Nifedipin-genericon pharma | Nifehexal;
- (AU) Australia: Adalat | Adapine | Addos xr | Adefin | Adefin xl | Apo-Nifedipine XR | Chemmart Nifedipine | Cm nifedipine | Nifecard | Nifedipine-bc | Nifehexal | Nyefax | Sbpa nifedipine | Terry White Chemists Nifedipine | Tw nifedipine;
- (BD) Bangladesh: Ampine | Fenamon | Ficard | Nefelon | Nicodin | Nidipine | Nifecap | Nifecard | Nifedin | Nificap | Nifin;
- (BE) Belgium: Adalat | Hypan | Nifedipine doc | Nifedipine eg | Nifedipine mylan | Nifedipine ratiopharm | Nifeslow;
- (BF) Burkina Faso: Adalat | Adalate la | Arconifed sr | Nifcal | Nifedi denk 10 retard | Nifedi denk 20 retard | Nifedipine lp tm;
- (BG) Bulgaria: Adalat | Adalat Eins | Adalat oros | Cordaflex | Cordipin xl | Corinfar | Korincare | Korincare neo | Nifedipin;
- (BR) Brazil: Adalat | Adalex | Adcor | Cardalin | Cronodipin | Dilaflux | Dilavax | Dipinal | Loncord | Neo fedipina | Nifadil | Nifedax | Nifedicard | Nifedipress | Nifehexal | Nioxil | Normopres | Oxcord;
- (CH) Switzerland: Adalat | Cardipin | Corotrend | Ecodipin | Nife-basan | Nifedipin genericon | Nifedipin Helvepharm | Nifedipin mepha | Nifedipin Sandoz | Nifedipin spirig | Nifedipin spirig hc | Nifedipin upsa;
- (CI) Côte d'Ivoire: Adalat la | Adalate | Apo nifedipine | Arconifed sr | Nifesha;
- (CL) Chile: Adalat | Angiodisten | Carbloc | Cardicon | Nifedipino | Nifedipino L.Ch. | Nipress | Nipress Oad | Pabalat | Sulotil;
- (CN) China: Adalat | De gao ning | Ecodipin | Er kang bi tong | Heng xin | Jiu bao ka di | Jiu bao ping | Jupocardia | Na xin tong | Ni fu da | Nifedipin ratiopharm | Nifedipine extended release tablets (iii) | Nifedipine sustained release tablets (i) | Nifedipine sustained release tablets (ii) | Nifedipine sustained release tablets (iv) | Sai yan teng | Sheng tong ping | Tian hai li | Xin ran | Yi xin | Yi xin ping;
- (CO) Colombia: Acerdil 24 horas | Beclazone | Cardiosol | Nicaloc | Nifecor | Nifedipina | Nifedipino | Nifetabs | Nifevas-O | Tensomax | Tensopin;
- (CR) Costa Rica: Nifedipina;
- (CZ) Czech Republic: Adalat | Apo nifed | Cordafen | Cordipin | Cordipin xl | Corinfar | Coronipin | Depin e | Nifecard | Nifehexal | Pidilat | Sponif | Supracordin;
- (DE) Germany: Adalat | Adalate | Aprical | Cisday | Coracten | Cordicant | Corinfar | Corotrend | Dignokonstant | Duranifin | Jedipin | Jutadilat | Nife | Nife biochemie | Nife klast | Nife wolff | Nife-puren | Nifeclair | Nifecor | Nifedipat | Nifedipin | Nifedipin al | Nifedipin denk | Nifedipin Sandoz | Nifedipin stada | Nifedipin Verla | Nifehexal | Nifelat | Nifical | Nifsigma | Pidilat;
- (DK) Denmark: Adalat oros | Nifedipin alternova;
- (DO) Dominican Republic: Adalat | Adalat oros | Cardiomed | Corapres | Dilcor | Evopril sr | Fusepina | Lufradipina | Nicardia xl | Nifecol | Nifed | Nifedicor | Nifedilat | Nifedip | Nifedipina | Nifedipina alfa | Nifedipina Inmenol | Nifedipina Lam | Nifedipina retard lam | Nifedipino | Nifedipres | Nifek | Nifelac | Nificard | Nifpin AP | Nitensor | Presnife retard 30 | Presnife retard 60 | Tensopin | Unifedipina;
- (EC) Ecuador: Nifedin | Nifedipina | Nifedipino | Nifelan | Zenusin;
- (EE) Estonia: Adalat | Adalat crono | Adalat oros | Cordafen | Cordaflex | Cordipin | Cordipin xl | Corinfar | Nifangin | Nifebene | Nifecard | Nifedeks | Nifedipin | Nifedipin ratiopharm | Nifedipina mylan | Nifehexal | Nifesan | Nifetard | Niphedipin | Nycopin | Nycopin depo | Phenihydin | Sanfidipin | Sponif;
- (EG) Egypt: Adalat | Cardiopine | Dilcor | Epilat | Epilat retard | Nifepin | Vasonipine;
- (ES) Spain: Adalat | Adalat oros | Cordilan | Dilcor | Nifedipino bayvit | Nifedipino ratiopharm | Pertensal;
- (ET) Ethiopia: Depin e retard | Epilat retard | Nifedi denk 10 retard | Nifedi denk 20 retard;
- (FI) Finland: Adalat | Adalat oros | Coronipin | Enilevol | Nical | Nifangin | Nifdemin | Nifecor | Nifedipin alternova | Onidin;
- (FR) France: Adalate | Chronadalate | Mapakna lp | Nifedipine Arrow | Nifedipine eg | Nifedipine

g gam | Nifedipine gnr | Nifedipine merck | Nifedipine ratiopharm | Nifedipine rpg | Nifedipine sandoz | Nifedipine teva | Nifedirex;

- (GB) United Kingdom: Adalat | Adanif | Adipine | Adipine mr | Angiopine | Calanif | Calchan mr | Calcilat | Cardilate mr | Cardiofin | Coracten | Coracten sr | Coroday | Dexipress MR 20 | Fortipine | Fortipine la | Fortipine nye | Genalat | Hypolar | Kentipine | Nidef | Nifedipine Arrow | Nifedipine berk | Nifedipine cox | Nifedipine kent | Nifedipine sandoz | Nifedipress | Nifedipress berk | Nifedipress cox | Nifedipress sandoz | Nifedotard | Nifelease | Nifensar | Nifopress | Nimodrel | Nivaten | Slofedipine | Tensipine MR | Unipine xl | Valni | Vasad;
- (GH) Ghana: Cardigen retard | Ndip | Ndip xr | Nifedose | NIFIN 20 R | Novodep | Pinek | Pinfed sr | Retardine XL | Vl nifedipine | Vynif;
- (GR) Greece: Adalat | Antiblut | Coracten | Flecor n | Glopir | Macorel | Nefelid | Nifedicor | Nifedipat | Nifedipin al | Nifelat | Viscard;
- (GT) Guatemala: Nifedi denk | Nifedipina;
- (HK) Hong Kong: Adalat | Apo nifedipine | Cardilate mr | Carditas | Coracten | Cordipin | Nadipinia | Nifecard | Nifecard xl | Nifelat | Nifopress | Vasdalat | Waridipin;
- (HR) Croatia: Cordipin | Nifedipin | Nifedipin Pliva;
- (HU) Hungary: Adalat | Cordaflex | Cordipin xl | Corinfar | Huma-nifedin | Nidipin | Nifecard xl | Nifedipin al;
- (ID) Indonesia: Adalat | Calcianta | Calcigard | Carvas | Cordalat | Coronipin | Farmalat | Fedipin | Ficor | Inficard | Nifecard | Nifedin | Pincard | Vasdalat | Vasoner | Xepalat | Zendalat;
- (IE) Ireland: Adalat | Adanif | Adefin xl | Coracten | Nifed | Nifelease | Nifensar | Systepin | Tensipine MR | Vasofed;
- (IL) Israel: Megalat | Nifedilong | Nifedipine teva | Pressolat;
- (IN) India: Adalat | Alnipine | Calbloc | Calcigard | Calnif | Cardipin | Cardovasc xl | Cardules | Depicor | Depin | Dolocard | Myogard | Myogard la | Nacten | Needin | Nepin | Nexidin | Nf | Nfd retard | Nicardia | Nicardia retard | Nifed | Nifedine | Nifelat | Niferil;
- (IT) Italy: Adalat | Adalat a.r. | Adalat crono | Citilat | Coral | Euxat | Fenidina | Nifedicor | Nifedicron | Nifedin | Nifedipina | Nifedipina doc | Nifedipina eg | Nifedipina mylan generica | Nifedipina provvisoria | Nifesal | Nipin;
- (JO) Jordan: Adalat | Coracten | Epilat | Fenamon | Myogard | Nifecard;
- (JP) Japan: Adalat | ADALAT CR | Allotop l sawai | Allotop l tsumura | Alonix s | Anpect | Atanaal mita | Atanaal sawai | Atanaal towa | Atenerate kaken | Atenerate l mita | Atenerate merck hoei | Atenerate tsuruhara | Cardiobren | Cardioluft l | Cardioluft l fuji | Casanmil | Casanmil s | Celebrate | Cobalat | Cobalate kobayashi kako | Corinael | Corodilate | Emaberin | Emaberin l | Herlat | Kepakuru l | Kisalart hexal | Kisalart nichiiko | Knoramin | Knoramin chemiphar | Lemar | Marivolon | Menoprizin l | Milfadin | Moderat | Nalicidin | Nerabole kotobuki | Nerabole zeria | Nifedinipe cr | Nifedipine cr | Nifedipine cr nichiiko | Nifedipine cr sanwa | Nifedipine cr sawai | Nifedipine cr towa | Nifedipine fujimoto | Nifedipine Kyorin Rimedio | Nifedipine l | Nifedipine l nichiiko | Nifedipine l sanwa | Nifedipine l sawai | Nifedipine l towa | Nifedipine l tsuruhara | Nifedipine merck hoei | Nifedipine mylan | Nifedipine nichiiko | Nifedipine sawai | Nifedipine teva | Nifedipine tsuruhara | Nifedipine yoshindo | Nifelanterin cr | Nifelantern | Nifelat | Nifelat l | Nifeplan L | Nifeslow | Nifpine | Nirena | Palpeat kyowa hakko | Palpeat pfizer | Ramitalate | Ramitalate taiyo | Ramitalate towa | Ramitalate-L | Ramitlalate l kayaku | Ronian | Sepamit | Sepamit r | Siopelmin l | Towarat | Towarat l;
- (KE) Kenya: Adalat | Adalat la | Calcigard retard | Carditas retard | Depin e retard | Epilat retard | Nefdin | Nefin er | Nepif retard | Nepin sr | Nicardia | Nicardin sr | Nifecard xl | Nifed retard | Nifedi denk | Nifelat r | Nifelat retard | Nifpine retard | Nipine sr | Zenusin 20 sr;
- (KR) Korea, Republic of: Adalat | Adalat oros | Adapine osmo | Caranta | Daehanneprin | Dongkoo nifedipine | Fedipine 24 | Hadipine | Hawon nifedipine | Medicoret | Nadipine | Neprin | Nifedin | Nifedine | Nifedipinaeuderma | Nifedix | Nifedsol | Nifel | Nifelan | Nifendal | Niferon | Niferon cr | Nipidin | Nipine | Niporin | Pidipin | Sanodipine | Union nifedipine;
- (KW) Kuwait: Adalat | Coracten | Zenusin;
- (LB) Lebanon: Adalat | Apo nifed | Dilcor | Nifedicor | Nifedip | Nifelat | Nifelat r | Niferetard | Nipin | Zenusin;
- (LT) Lithuania: Adalat | Adalat oros | Apo nifed | Calcigarde | Cordafen | Cordaflex | Cordipin | Corinfar | Coronipin | Depin e | Nifadil | Nifebene | Nifecard | Nifedeks | Nifedipin | Nifedipin al |

Nifedipin ratiopharm | Nifedipin stada | Nifedipin stada uno | Nifehexal | Nifesan | Phenihidin | Sanfidipin;
- (LU) Luxembourg: Adalat | Aprical | Hypan | Nifedipine mylan | Nifehexal;
- (LV) Latvia: Adalat | Adalat oros | Calcigarde | Cordafen | Cordipin | Corinfar | Coronipin | Depin e | Duranifin | Fenamon | Nifadil | Nifecard | Nifedex | Nifedipin | Nifesan | Night rest | Phenihidin | Sanfidipin;
- (MA) Morocco: Adalate | Apo nifed | Chronadalate | Coracten | Nifedicor | Nifegen | Pharmadipine | Servidipine;
- (MX) Mexico: Adalat | Anhiten a | Apo-fidipisal | Cordilat | Corogal | Corotrend | Difepar | Fusepina | Gelprim | Kabloc | Linam | Nifar | Nifar gb | Nifedigel | Nifedipino | Nifedipino bruluart | Nifedipino gi | Nifedipino gi serr | Nifedipino hispanoamericano | Nifedipino protein | Nifezzard | Nifser | Notinzalten | Noviken LP | Pidef;
- (MY) Malaysia: Adalat | Adifen | Calcigard | Cordipin | Fenamon | Nicardia xl | Nifecard | Nifecip | Nifedipin | Nifehexal | Nifehexal retard | Nifelat | Niferin sr | Nudipin | Servidipine | Uphadipine;
- (NG) Nigeria: Acefex | Adalat la | Adopine | Cardipin | D glopa nifedipine | Eglofedipine | Ep-nif retard | Eufedipine | Exus nifedipine | Fladipine retard | Glopir retard | Hartipin | Jamcelia nifedipine | Lifeback nifedipine retard | Macfedipine | Milton nifedipine | Myolax | Nicardia xl | Nifebond | Nifecard xl | Nifecure | Nifedin dexcel | Nifetab | Normodipin | Orfedipine | Patuo sr | Rebok nifedipine | Replek nifedipine retard | Scrip nifedipine | Topnife | Transglobe nifedipine | Zenusin 20 sr | Zibadipin;
- (NL) Netherlands: Adalat | Adalat oros | Adalate | Nifedipin abz | Nifedipin acis | Nifedipin al | Nifedipin al t retard | Nifedipine A | Nifedipine accord | Nifedipine merck | Nifedipine PCH | Nifedipine ratiopharm | Nifedipine sandoz;
- (NO) Norway: Adalat | ADALAT CR | Adalat crono | Adalat la | Adalat oros | Adalat unimedic | Adalat xl | Nifedipin | Nifedipin al | Nifedipin ratiopharm | Nifedipine cf | Nifedipine sandoz | Nifenova | Nycopin;
- (NZ) New Zealand: Adalat | Adefin xl | Arrow nifedipine | Nyefax | Tensipine MR;
- (PA) Panama: Nifedi denk;
- (PE) Peru: Angidex | Angidex nf | Fediral | Nifedipino | Nifedipino genfar | Nifensar | Nifepharm | Tensomax | Unipine xl;
- (PH) Philippines: Adalat | Angipine | Calcibloc | Calcicor | Calgina | Cardicap | Darat | Denkifed | Fedcor | Fedipin | Hartigard | Heblopin | Heblopine | Hyperten | Nelapine | Nicardia | Nicardia xl | Nifecar | Nifedipine Medicap | Nifedipine Pacific | Nifelan | Nifestad | Normadil | Odipin | Servidipine | Tensibloc;
- (PK) Pakistan: Adalat | Anifed | Caranta | Cardipine | Masdipine sr | Nidipine | Nifecard | Nifed | Nifedi denk | Nifedicor | Nifedil | Nifelat | Ronian;
- (PL) Poland: Adalat | Cordafen | Cordipin | Corinfar | Nifecard | Nifedicor | Nifedipin acis | Nifedipin al | Nifedipin ratiopharm | Nifehexal;
- (PR) Puerto Rico: Adalat | Afeditab CR | Nifediac CC | Nifedical | Nifedipine er | Nifedipine extended release | Procardia;
- (PT) Portugal: Adalat | Adalat a.p. | Besdipina | Corinfar | Nifedate | Nifedipina | Nifedipina Generis | Nifedipina ratiopharm | Zenusin 20 sr;
- (PY) Paraguay: Adalat | Cuorex retard | Nifedil a.r. | Nifeten | Nifeten retard | Tensimeg | Tensoprel retard;
- (QA) Qatar: Adalat | Adalat LA | Adalat Retard | Adalat XL | Myogard | Nifedipin AL | Nifedipine AL | Nifedipress MR | Valni Retard;
- (RO) Romania: Adalat | Cordafen | Cordipin | Corinfar | Epilat retard;
- (RU) Russian Federation: Adalat | Calcigard | Calcigard retard | Calcigarde | Carinfer | Cordafen | Cordaflex | Cordipin | Cordipin xl | Corinfar | Corinfar retard | Fenamon | Nicardia | Nicardia cd retard | Nifebene | Nifecard | Nifecard xl | Nifedex | Nifedicap | Nifedipin | Nifedipin ns | Nifedipin-fpo | Nifedipine akrihin | Nifedipine mik | Nifehexal | Nifelat | Nifesan | Nificard | Osmo adalat | Phenihidin | Vero nifedipin;
- (SA) Saudi Arabia: Adalat | Apo nifedipine | Epilat | Nifecard;
- (SE) Sweden: Adalat | Adalat oros;
- (SG) Singapore: Adalat | Apo nifed | Cardiolat | Cordipin | Cordipin xl | Depin e retard | Fenamon | Nifecard | Nifedi denk | Nifedipine la | Nipin | Vasdalat;

- (SI) Slovenia: Adalat | Adalat oros | Cordipin | Nifecard;
- (SK) Slovakia: Cordafen | Cordipin | Cordipin xl | Corinfar | Nifecard | Nifehexal | Sponif;
- (SR) Suriname: Adalat | Adalat oros | Adalat retard | Apo nifedipine | Nifedi denk | Nifedipine sr | Nifelat | Nifelat r | Nifelat retard | Waridipin;
- (SV) El Salvador: Nifedipino;
- (TH) Thailand: Adalat | Adipine | Apo nifed | Calcigard | Carnif | Coracten | Depin e retard | Fenamon | Jedipin | Nelapine | Nicaloc | Nicardia | Nicardia cd | Nifecard | Nifedi denk | Nifedipin stada | Nifedipine ratiopharm | Nifelat | Nifelat r | Nifelat-q | Nificard | Nifiran | Niufedepine | Nyefax | Servidipine | Stada Uno | Towarat | Zenusin;
- (TN) Tunisia: Adalate | Anifed | Coracten | Corinfar | Nifedicor | Nifepine;
- (TR) Turkey: Adalat | Adalat crono | Kardilat | Nidicard | Nidilat;
- (TW) Taiwan: Adalat | Adalat cc | Adalat oros | Adapine | Ajulate | Alat | Alonix s | Apo nifed | Atanaal | Badipine | Cardiobren | Citilat | Coracten | Coral | Coronipin | Dekalat | Emaberin | Fedipine | Fenamon | Glopir | Harwell | Memoprizin | Nedipin | Neocardia | Nidepin | Nidomate | Nifate | Nifecardia | Nifed | Nifedin | Nifehexal | Nifelan | Nifelat | Nifepine | Niferos oros | Nirena | Nisimlone | Nisipin | Nit | Palpeat | Posipin | Ronian | Servidipine | Sidalat | Sindipine | Towarat;
- (UA) Ukraine: Adalat | Cordafen | Cordaflex | Cordipin | Cordipin retard | Cordipin xl | Corinfar | Corinfar retard | Epilat | Fenamon | Nicardia | Nifecard | Nifedicap | Nifedipin | Nifelat | Osmo adalat | Phenihidin;
- (UG) Uganda: Adalat la | Agopine | Calcigard retard | Carditas retard | Fenamon sr | Nepin sr | Nicardia | Nifedi denk 20 retard | Nifelat | Nifelat retard;
- (UY) Uruguay: Adalat | Ciruton | Cordipin | Cuorex | Cuorex retard | Ficarel | Nifed sol | Nifedel | Nifedilat | Nifedin | Nifedipina | Novodipen | Tonacor;
- (VE) Venezuela, Bolivarian Republic of: Adalat | Adalat oros | Biodipina | Cardiosol | Conducil | Fedilex | Nifal | Nifebiopina | Nifedicor retard | Nifedipina | Nifedipina lp | Nifedipino | Nioxil | Tensodiam | Tensomax | Tensopin;
- (VN) Viet Nam: Nifedipin hasan 20 retard | Nifedipin t20 retard | Nifehexal 30 la | Nifephabaco | Pymenife;
- (ZA) South Africa: Adalat | Anginor-20 cr | Apo nifedipine | Bio nifedipine xl | Cardifen | Cardilat | Cipalat | Fedaloc | Fidocare xl | Gitsalat | Macorel | Nifedalat | Norton-nifedipine | Rolab-nifedipine | Spec nifedipine | Vascard;
- (ZM) Zambia: Calcigard | Calnif | Cardifen | Carditas retard | Nifedi denk | Nifelat | Nipine sr;
- (ZW) Zimbabwe: Adalat xl | Calcigard retard | Nifelat

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